

Value and Benefits of an Internet eXchange Point (IXP)



Bangladesh

Agenda

01 What is an IXP

02 Short and Long Term Benefits of an IXP

03 Increasing the IXP Value



Key Milestones in Internet Development

Early Internet Introduction

Internet introduced in early 1990's, starting the country's digital journey. First ISPs launched in 1996 expanding access.

Growth of Internet

Late 1990s–early 2000s: Rapid growth of licensed ISPs. Connection to SEA-ME-WE-4 submarine cable in 2006

Mobile Broadband Expansion

Mobile broadband introduced in 2013, greatly increasing internet access especially in rural regions.

Cybersecurity and 4G Launch

2008 saw creation of BD-CERT for cybersecurity; 2018 launched 4G and mass adoption.



Key Milestones in Internet Development

 Bangladesh

Nearby: [China](#) · [India](#) · [Myanmar](#) · [Nepal](#) · [Bhutan](#)

Summary

ASNs	1,958
IPs	1,997,775 (15% pingable)
Router IPs	7,358
Hosting IPs	100,529
Mobile Carriers	4
VPN Providers	1
Registry	APNIC



International Internet Routes

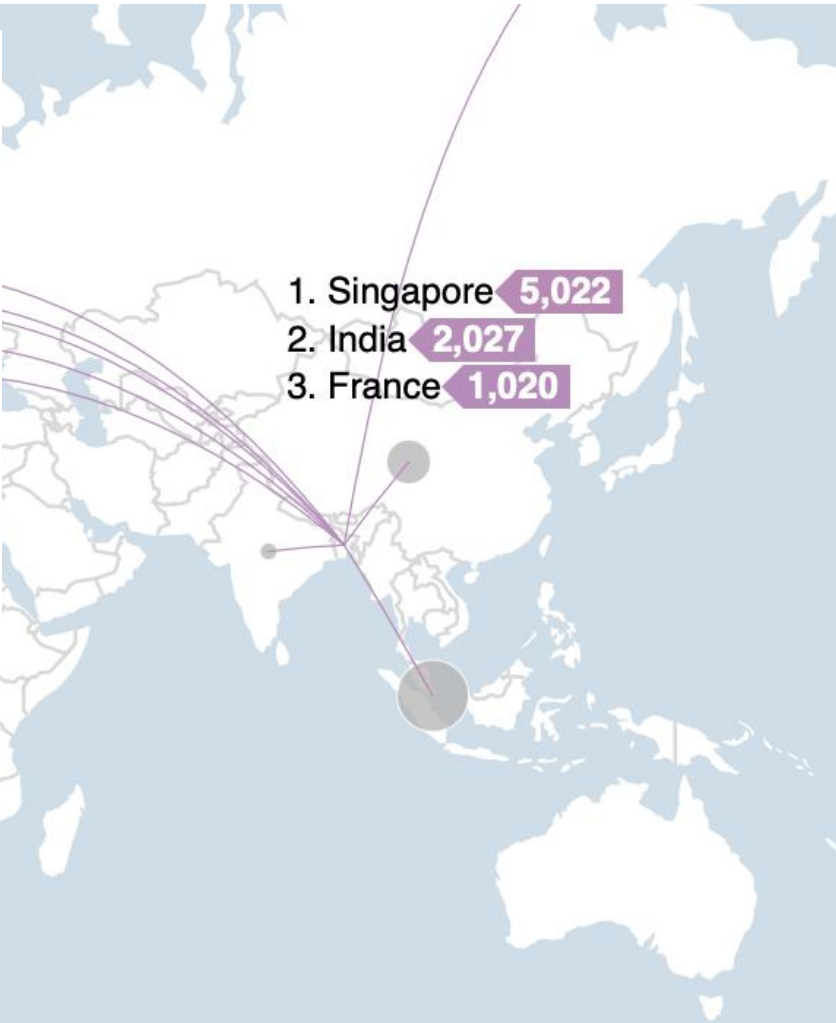


International Internet Routes

Rank	Route	2025 Bandwidth
1	Bangladesh – Singapore	5,022
2	Bangladesh – India	2,027
3	Bangladesh – France	1,020
4	Bangladesh – United Kingdom	74
5	Bangladesh – China	40
6	Bangladesh – Germany	40
7	Bangladesh – Italy	20
8	Bangladesh – Netherlands	20
9	Bangladesh – United States	12

International Internet Bandwidth (Gbps)

Country	2021	2022	2023	2024	2025
Singapore	1,786.0	2,386.0	3,312.0	4,125.0	5,022.0
India	632.0	892.0	1,122.0	1,532.0	2,027.0
France	260.0	410.0	720.0	920.0	1,020.0
United Kingdom	42.0	42.0	74.0	74.0	74.0
China	20.0	20.0	40.0	40.0	40.0
Germany	20.0	20.0	40.0	40.0	40.0
Italy	10.0	10.0	20.0	20.0	20.0
Netherlands	10.0	10.0	20.0	20.0	20.0
United States	6.0	6.0	12.0	12.0	12.0
Total	2,786.0	3,796.0	5,360.0	6,783.0	8,275.0



WHAT IS AN IXP



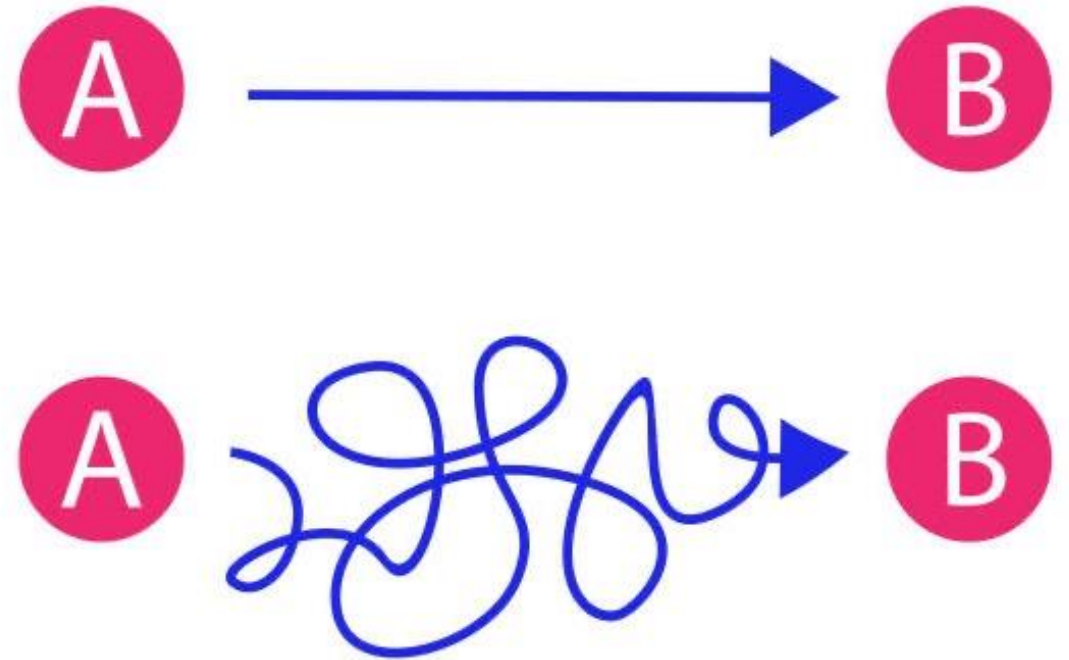
Photo: Jason Thien's photo, licensed as CC BY 2.0

Internet Exchange Points

The fastest route from Point A to Point B is a straight line, but when Internet infrastructure is underdeveloped, sometimes data zigzags through faraway places. The result is that Internet access is slower and more expensive.

An Internet Exchange Point (IXP) is a physical location where Internet infrastructure companies like ISPs and CDNs connect with each other.

IXPs create shorter, more direct routes for Internet traffic, making Internet access faster, cheaper, and more reliable.



Internet eXchange Points (IXP) – Bangladesh

Bangladesh

IXP Tracker

There are **8** active Internet Exchange Points (IXPs) in Bangladesh, with a combined total of **176** members as of **May 2026**.

IXPs are present in **1** of the **17** population centers over 300,000. In general, when networks use an IXP rather than direct transit, it reduces the reliance on international traffic. This can help keep parts of the Internet available, even during disruptions or outages.



Region

Asia

Sub-region

Southern Asia

Population

173,562,364

UN classification(s)

Least Developed Country

Active Networks

1520

Explore by country:

Bangladesh

View country

Find an IXP:

Select an IXP or filter by country

View IXP

8

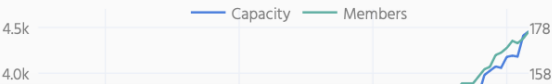
Total number of IXPs, as listed in PeeringDB. Last updated May 2026.

[View IXP list](#)

IXP membership and capacity

One way to measure the strength of IXPs in the Internet landscape is to look at membership and capacity. Larger networks tend to bring more capacity, which can benefit all members.

Measures changes in membership and capacity across all IXPs in **Bangladesh**, as shown in PeeringDB as of May 2026.



Domestic network coverage at IXPs in Bangladesh



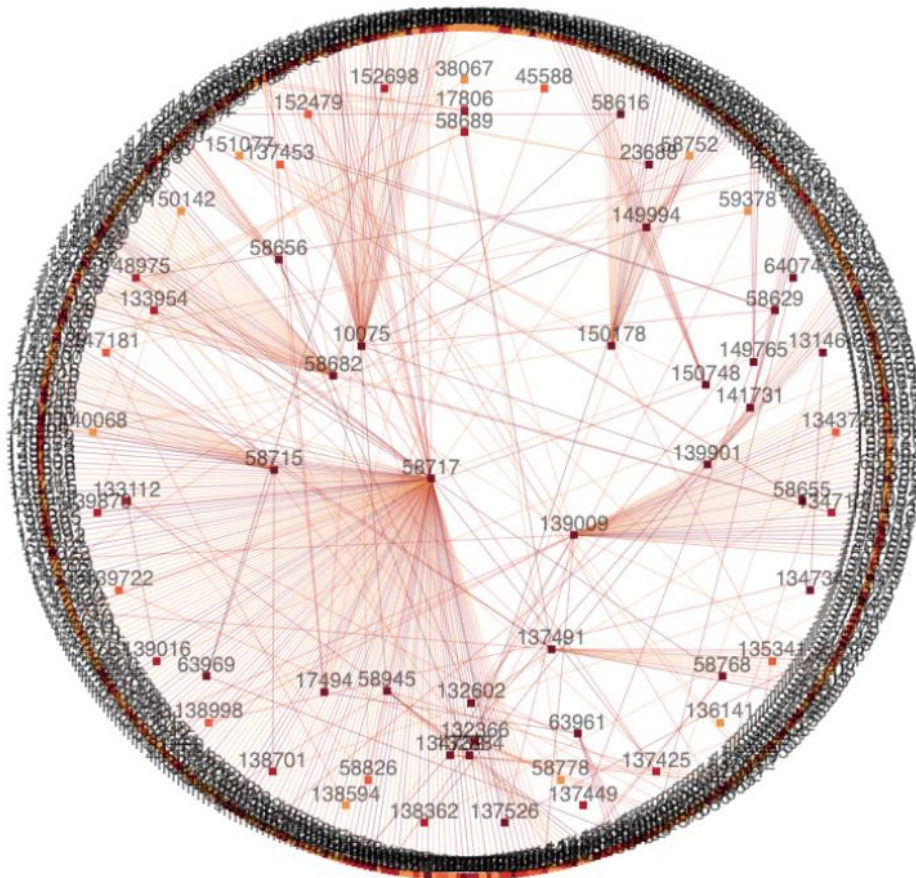
https://pulse.internetsociety.org/en/ixp-tracker/?country_code=BD

Autonomous System Interconnection – Laos

Visible IPv6 interconnections

470 interconnections among
495 Autonomous Systems

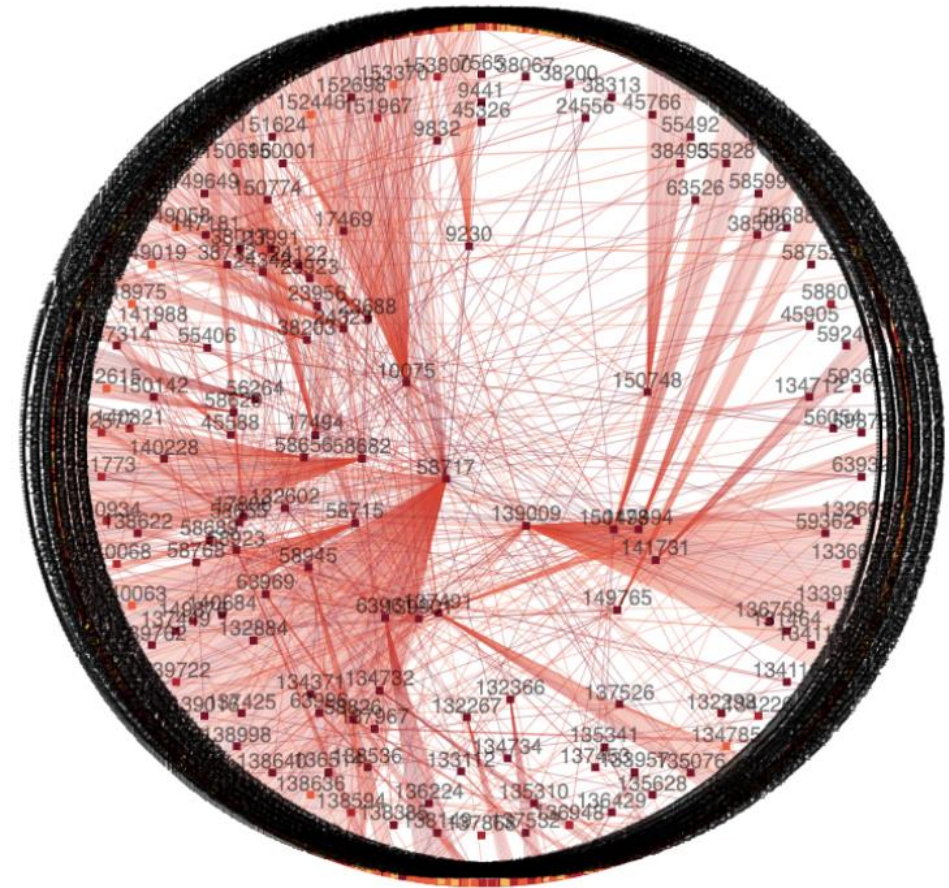
Q Search ASN here



Visible IPv4 interconnections

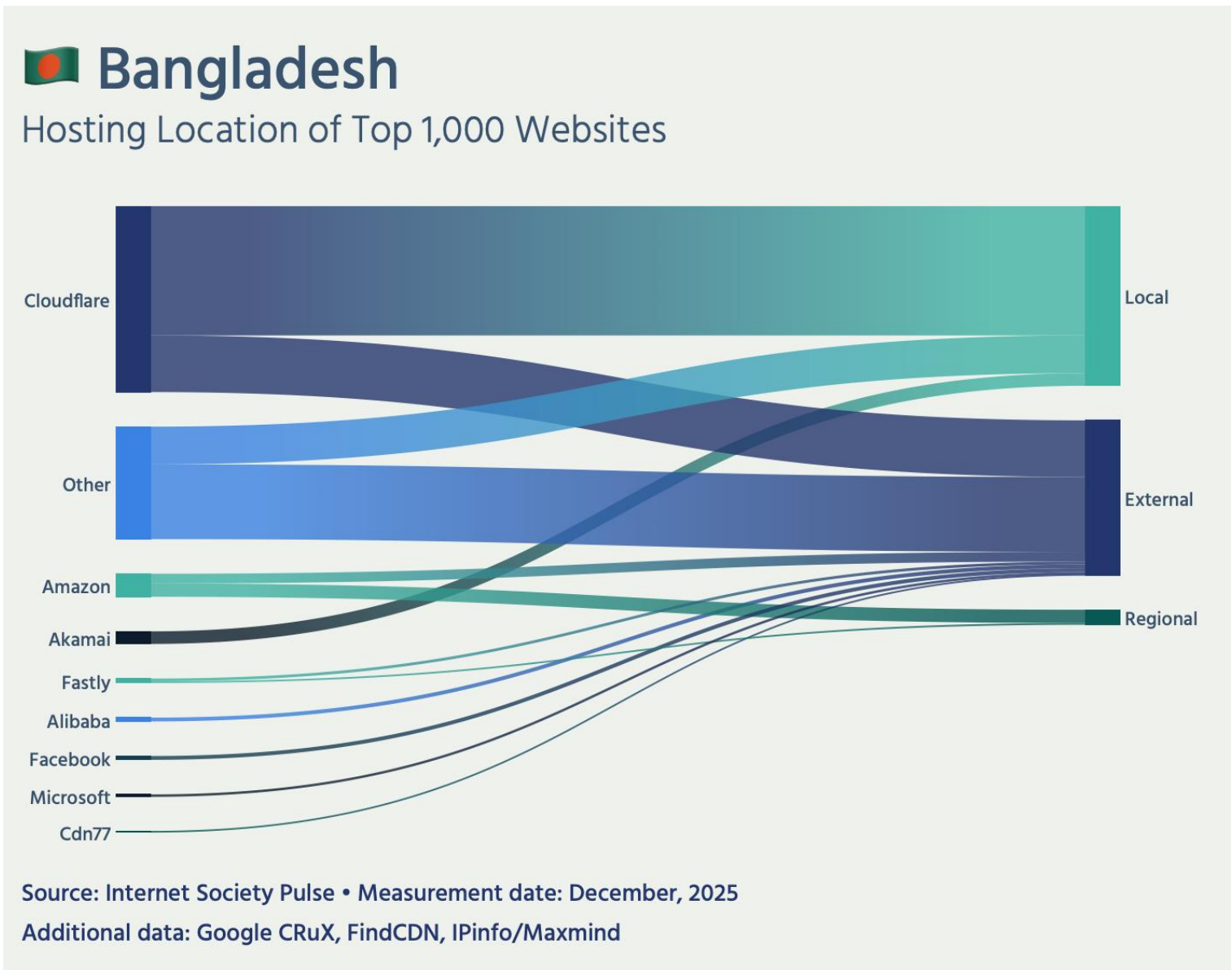
1602 interconnections among
1509 Autonomous Systems

Q Search ASN here



<https://rex.apnic.net/as-interconnections?allocationType=ipv4,ipv6&economy=BD>

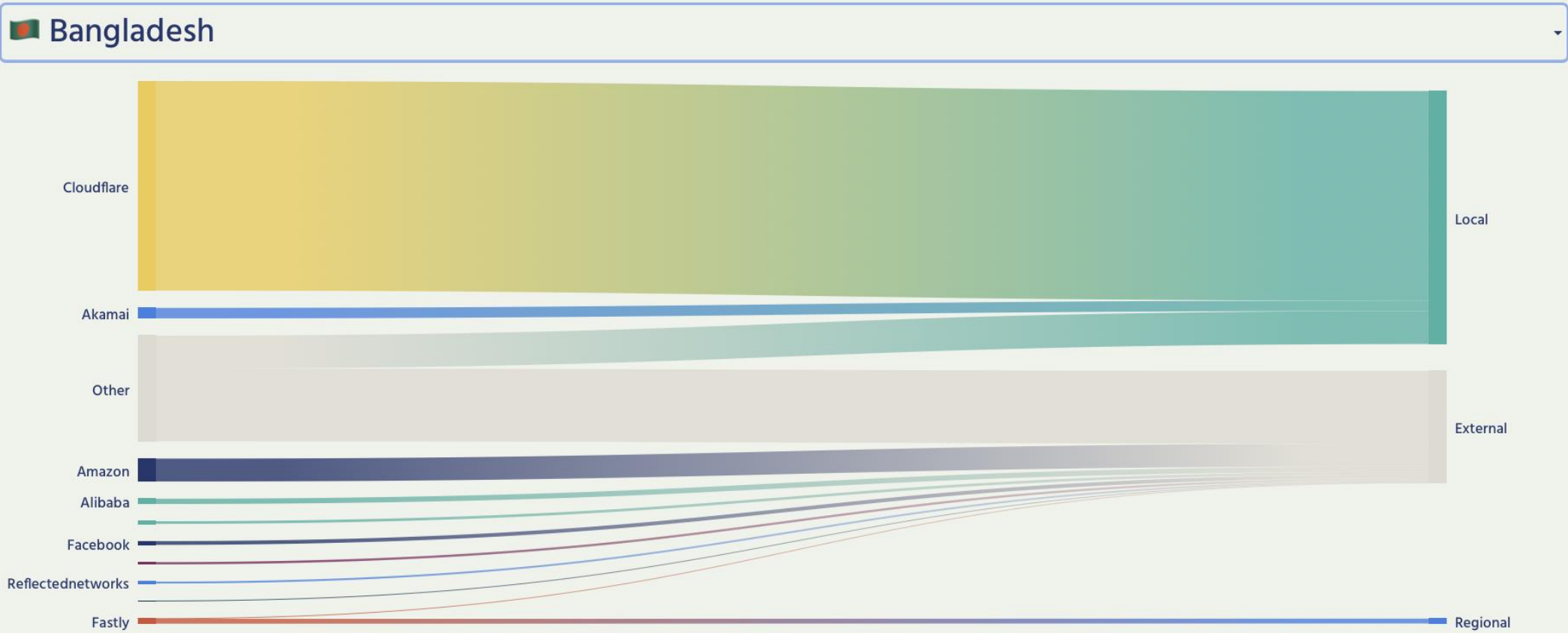
Data Example – Bangladesh



Data Example – Bangladesh

Locally Cached Content

Hosting Location of Top 1,000 Websites



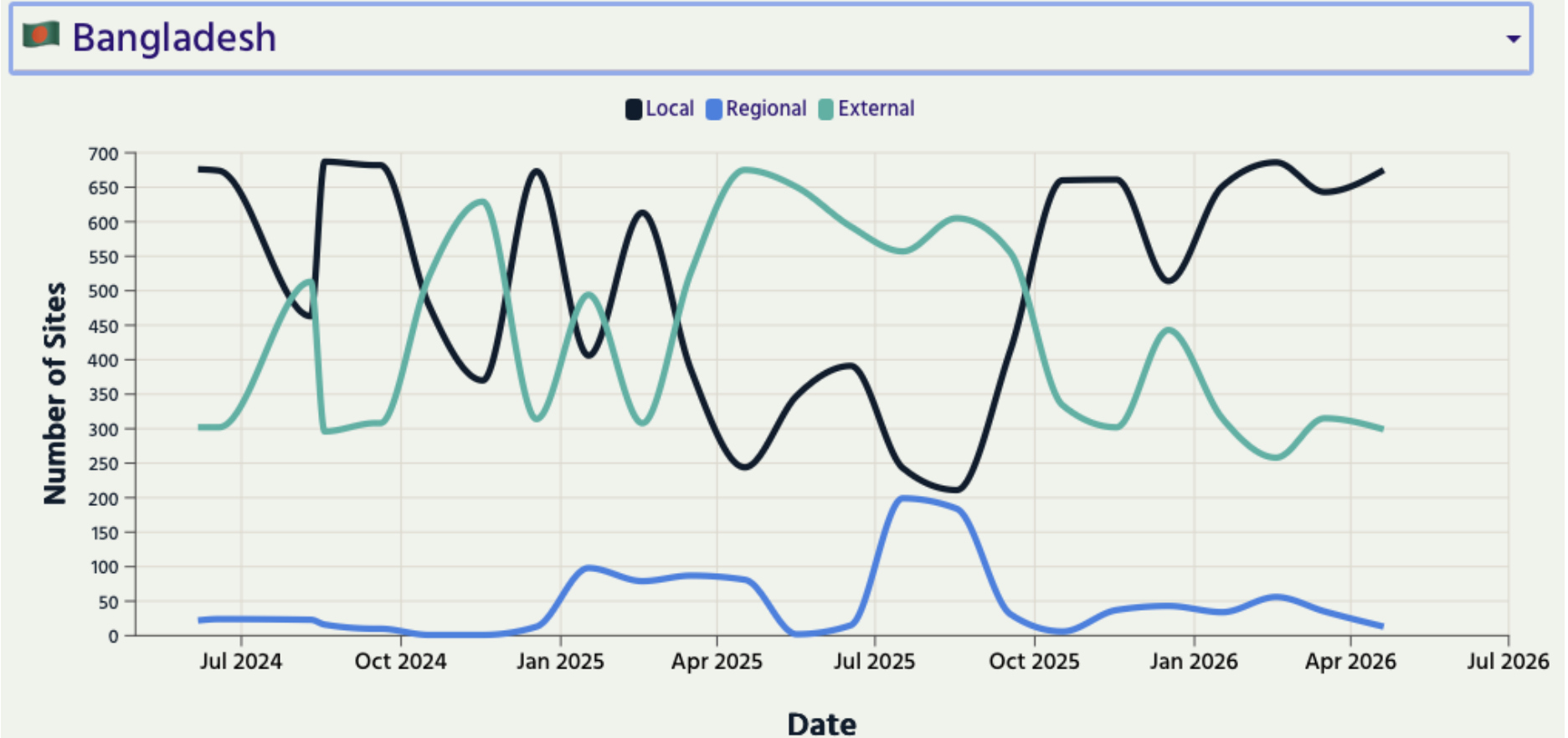
Source: [Internet Society Pulse](#) • Measurements updated every month
Additional data: Google CRuX, FindCDN, IPinfo/Maxmind



Data Example – Bangladesh

Building the Local Network

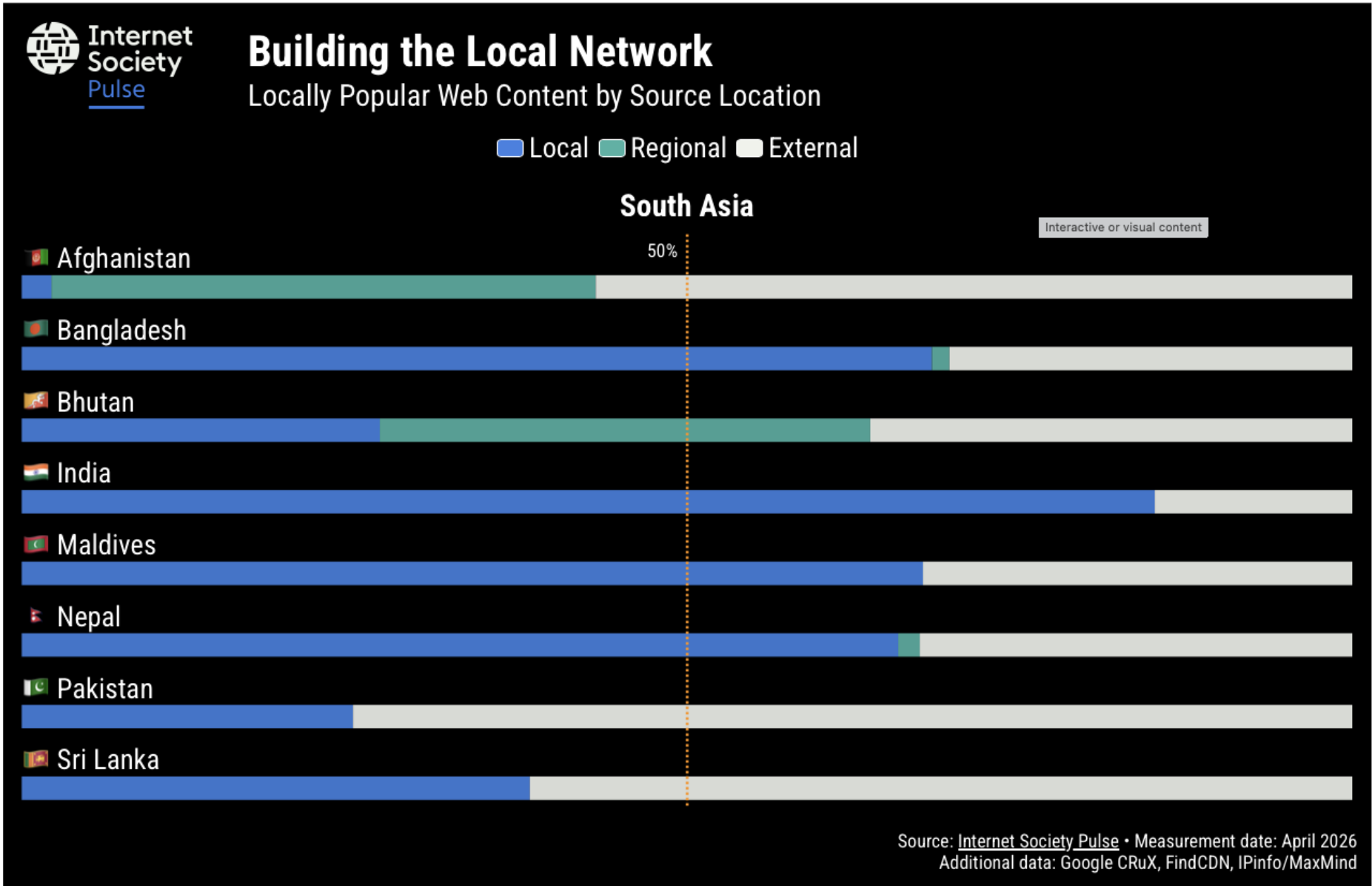
Popular content locality measurements over time



Source: [Internet Society Pulse](#)

Data: Google CRuX, FindCDN, IPinfo/MaxMind

Data Example – Bangladesh



IXP Case Study – Maldives (MVIX)

The Maldives IXP (MVIX) started operating in 2022. This is a huge success of our mutual multistakeholder efforts to establish neutral and open IP peering facilities for network operators/Internet service providers in small island developing states (SIDS).

Before the IXP, all local data was accessed and transferred via international transit peers, which was expensive and slow.

Keeping traffic local instead of sending it via international routes provides better resilience, stability, efficiency, and quality improvements all at a lower cost.



MVIX Impact

- Right from the start, MVIX's impact on increasing local traffic and reducing reliance on expensive transit links was clear.
- In the months after starting operation, Maldives IX (MVIX) has recorded a massive increase in local Internet traffic from 700mbps to 2.5Gbps (280% growth).
- This is also thanks to a primary CDN (e.g., Meta, Cloudflare) peering when the IXP start operating

Popular Content Locality

A measure of how much locally popular web content is hosted in-country or in-region.

70%

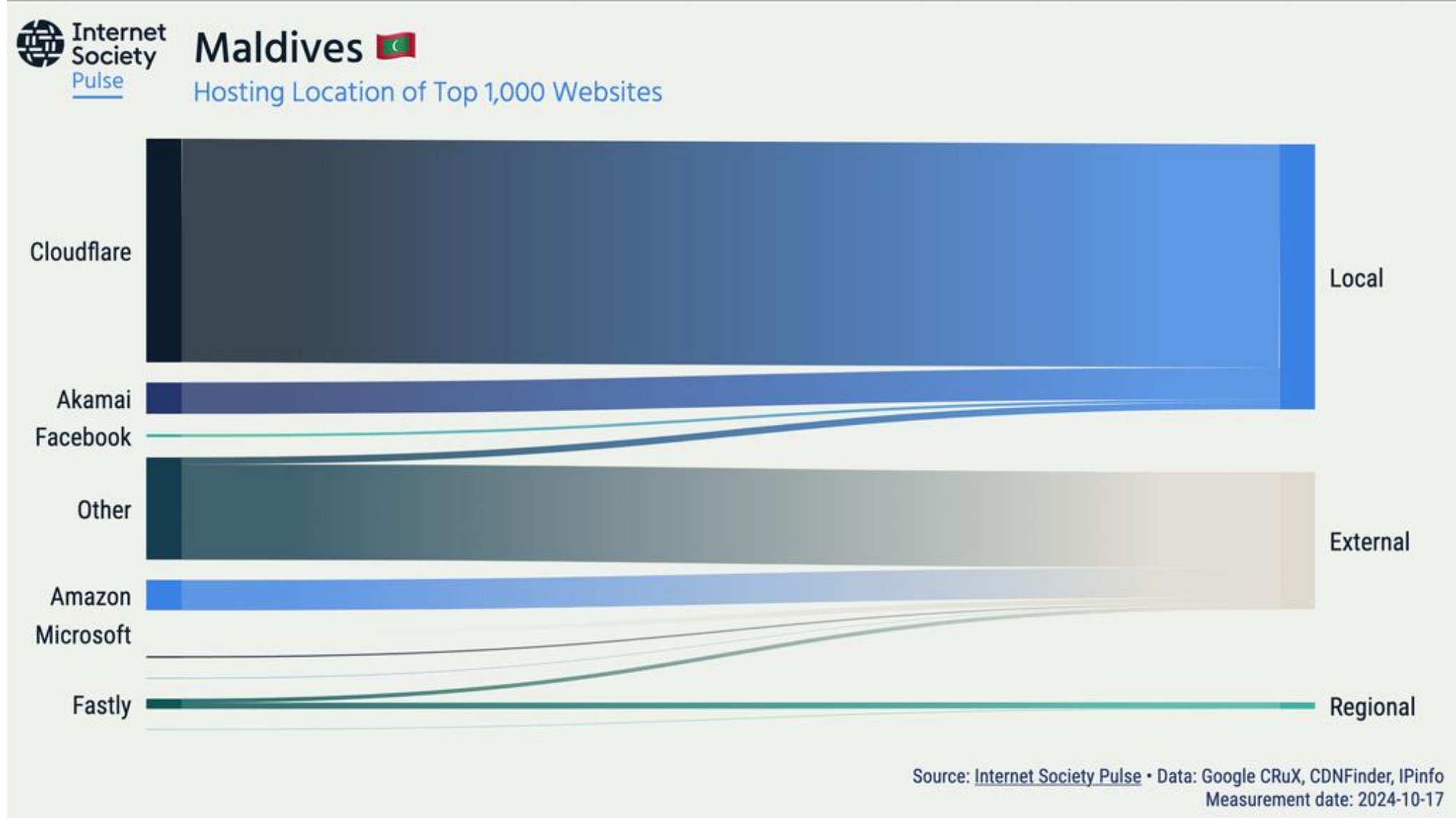
Regional Rank: 12th

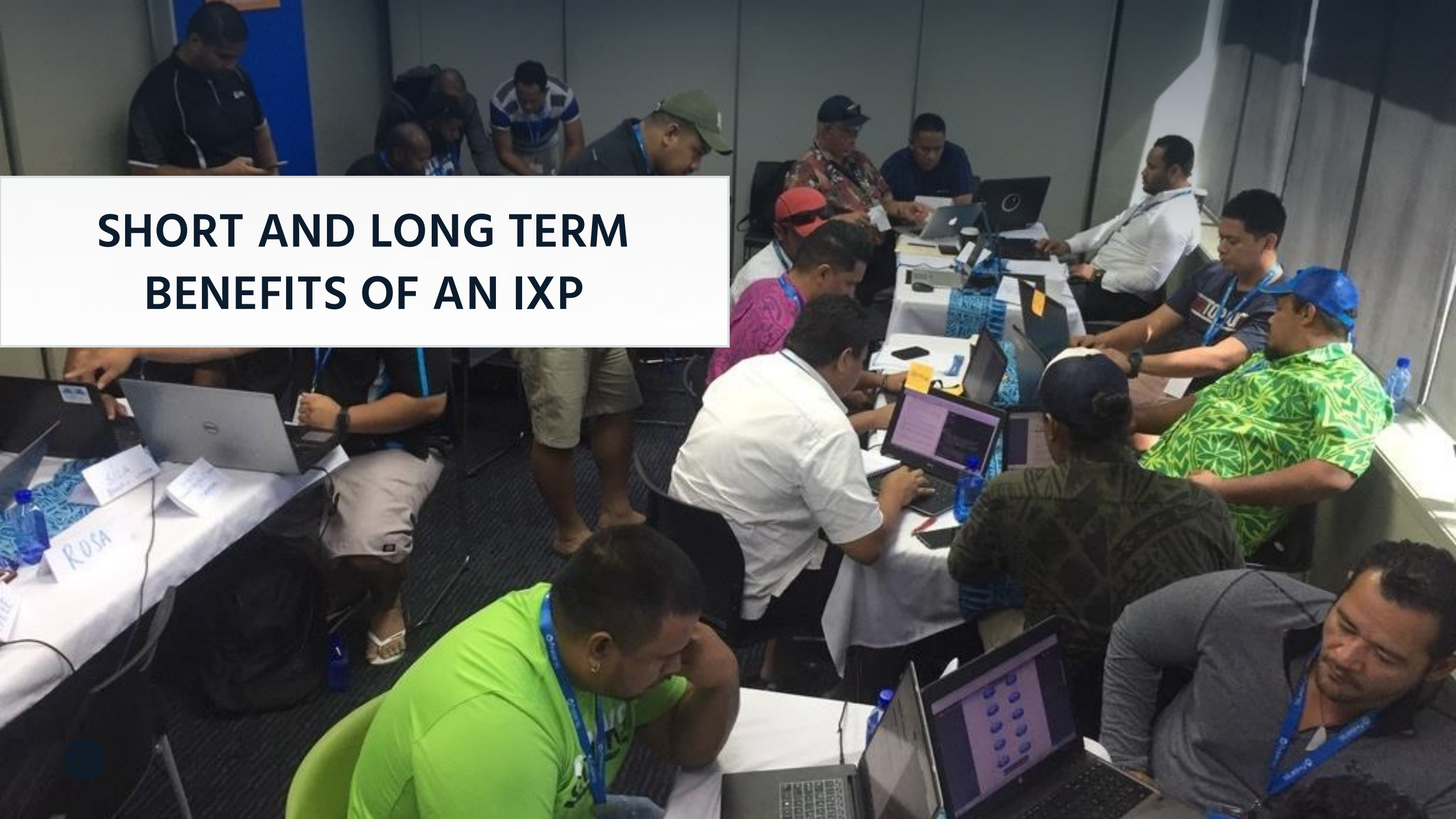
Asia average: 48%

[See c](#)



Popular Content Locality - CDN distribution



A group of people are seated around a long table in a meeting room, working on laptops. The room has large windows in the background. The people are wearing casual attire, including t-shirts, button-down shirts, and caps. Some are looking at their laptops, while others are looking towards the camera. The text "SHORT AND LONG TERM BENEFITS OF AN IXP" is overlaid on the image in a white box.

SHORT AND LONG TERM BENEFITS OF AN IXP



Benefits of IXPs: Short Term



Significant cost savings by avoiding expensive long-distance data routing.



Local users gain more bandwidth due to lower local capacity costs.



Local traffic is up to 10x faster due to reduced latency and fewer hops.



Connecting to an IXP requires BGP knowledge and IP resources (IP addresses, ASNs), plus skilled engineers.



Greater autonomy: IXPs reduce reliance on international links, improving local internet stability.



Benefits of IXPs: Long Term



Enables growth of carrier-neutral data centres and local hosting markets.

Easier access to local content (e.g., government e-services) encourages content creation.

Supports new local services needing fast, affordable connections and broad reach.

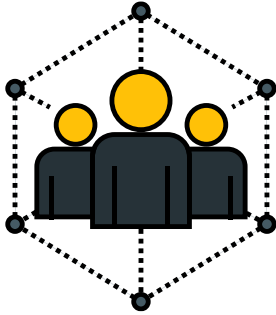
Offers more upstream provider options, boosting competition in the transit market.



INCREASING THE IXP VALUE



IXP Business Model



Importance of Business Model

- Voluntary support isn't scalable; a solid model ensures service quality.
- Enables hiring staff, maintaining equipment, and supporting growth.
- Builds trust with members and supports activities like training workshops.



Developing IXP Business Model

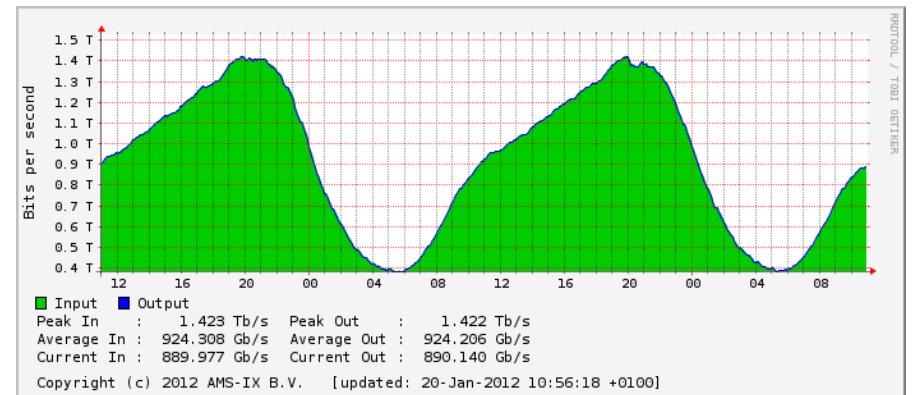
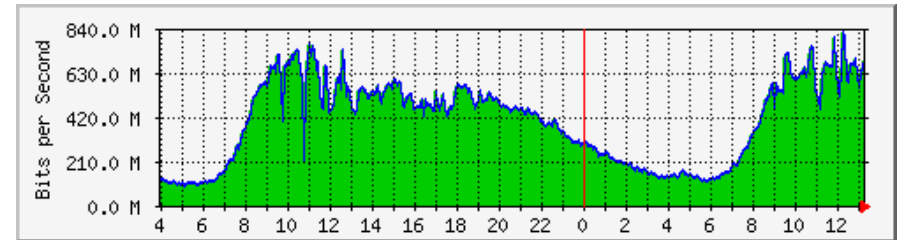
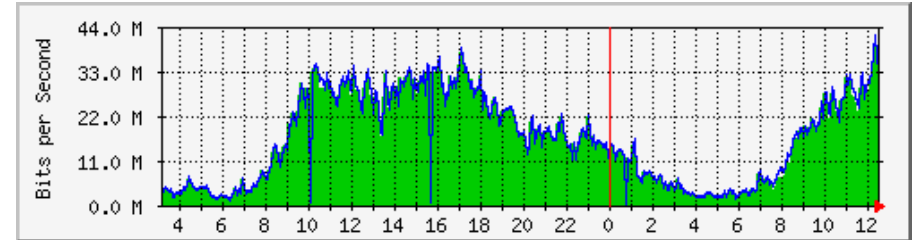
- Based on member agreements and primarily funded through port fees.
- Startups often charge a flat fee per port, regardless of speed.
- As demand grows, pricing by port speed helps fund capacity upgrades.



Value Add: Growing Critical Mass

Opening an IXP to non-traditional members: Which IXP would you prefer to Peer?

- E- Government Services such as Customs, immigration, Examinations, etc.
- National Research and Education Network (NRENs)
- Country Code Top Level Domain (ccTLD) Registries
- Content Providers such as, IP-TV providers, VoIP providers, etc
- Mobile network providers offering 2G/3G and LTE services
- Banking for online banking Services



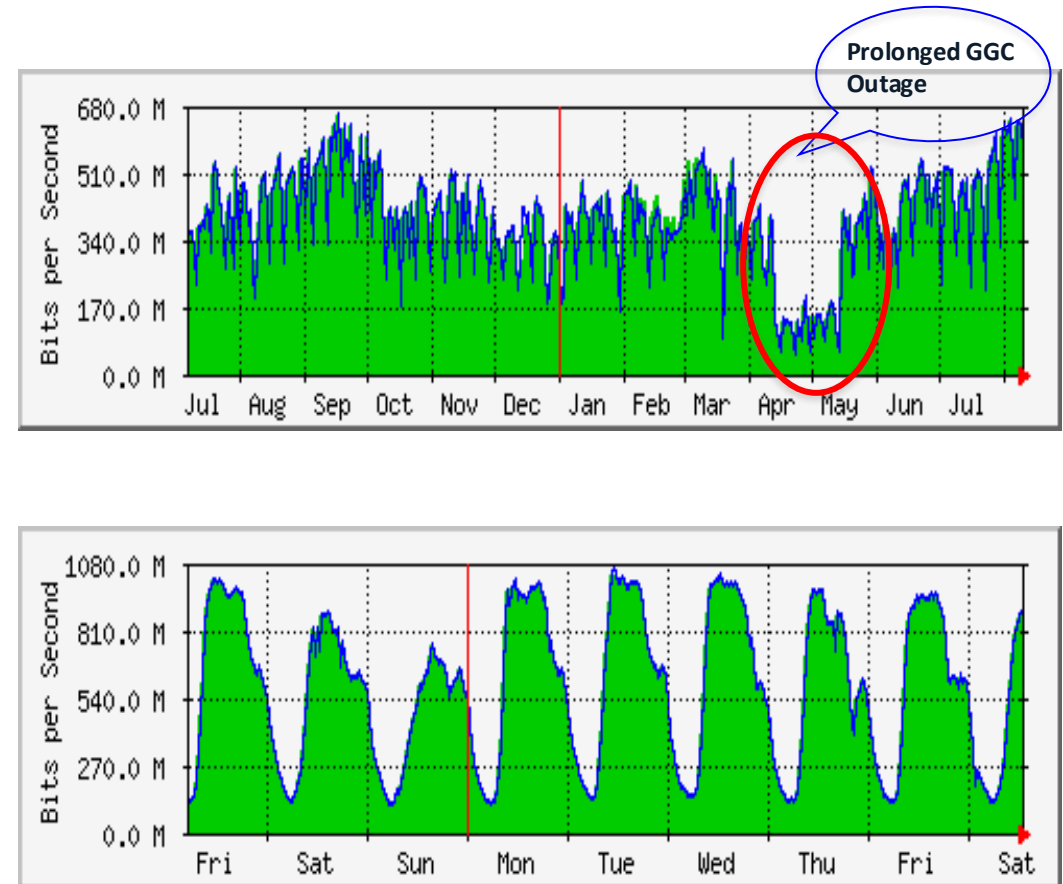
Value Add: CDN Edge Caching

Value of CDN Edge Caching

- Content Distribution Networks (CDN) aim to deliver your content to visitors as quickly as possible
- Edge Caching takes this concept one step further by geographically placing servers closer to users.
- Edge caches are often placed close to IXPs to target the large concentration of user networks (or eyeball networks)
- Google's Global Cache (GGC) program is one of the successful programs that IXPs can consider to implement in partnership with their members



Impact of CDN Cache at KIXP



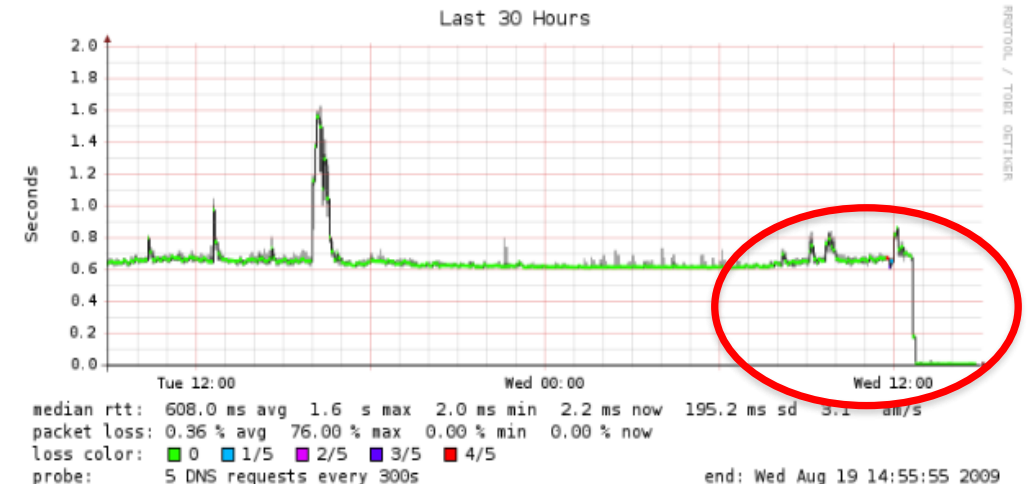
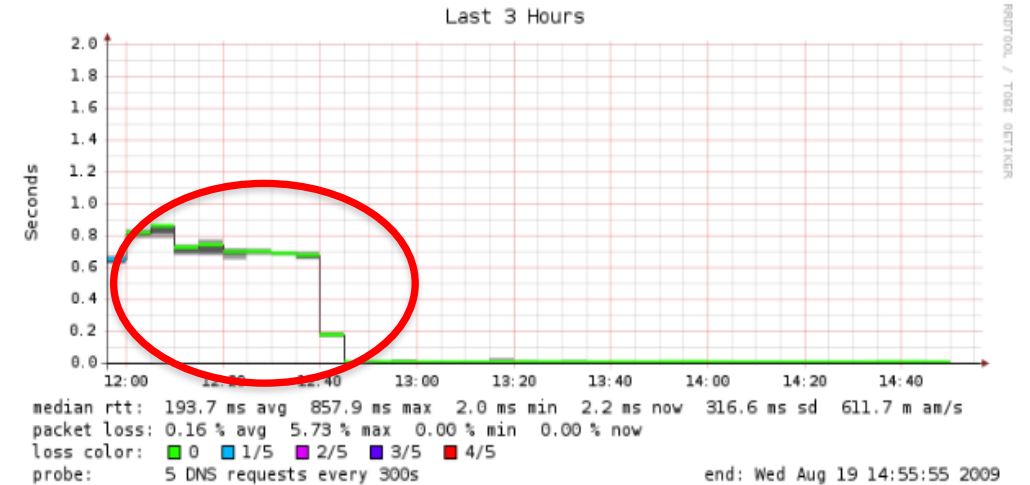
Value Add: Complementing Services

Additional services that increase value of the IXP are:

- **Domain Name System (DNS) Root Server Instances**
 - DNS Root servers instances at an IXP help improve the resilience of the local internet.
- **Network Time Server**
 - Implementing a stratum 1 clock using GPS is a simple and effective way of providing NTP services at the IXP.
- **Looking Glass**
 - Looking glass is an important troubleshooting tool. It also collects data that is of interest to researchers.
- **Route Server**
 - Simplifies the setup for most new entrants at an IXP.
- **Routing Registry**
 - IXP Routing Registry is a useful tool that assists members configure their IXP interconnection.

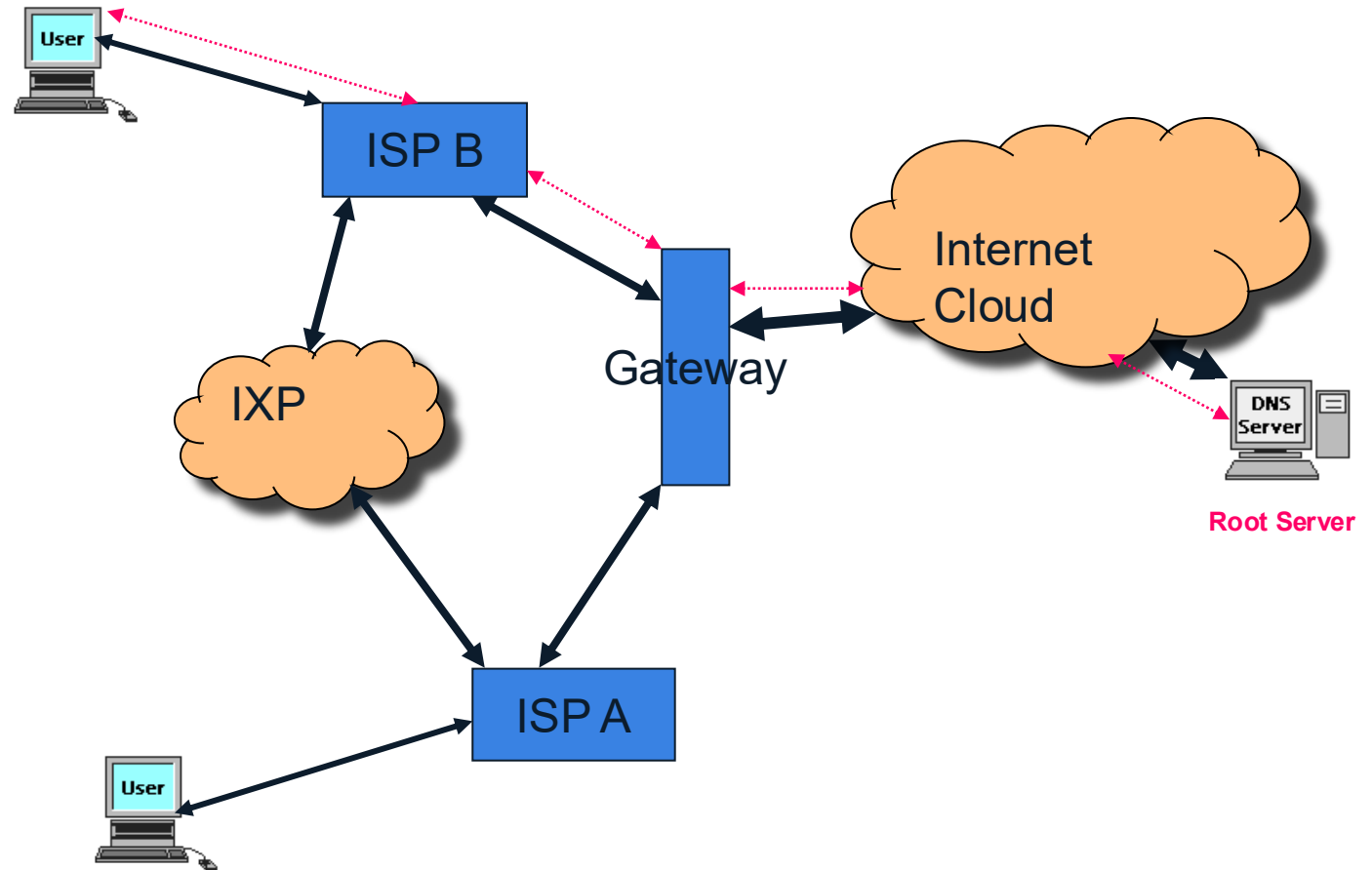


Impact of K-Root Server Instance at TIXP

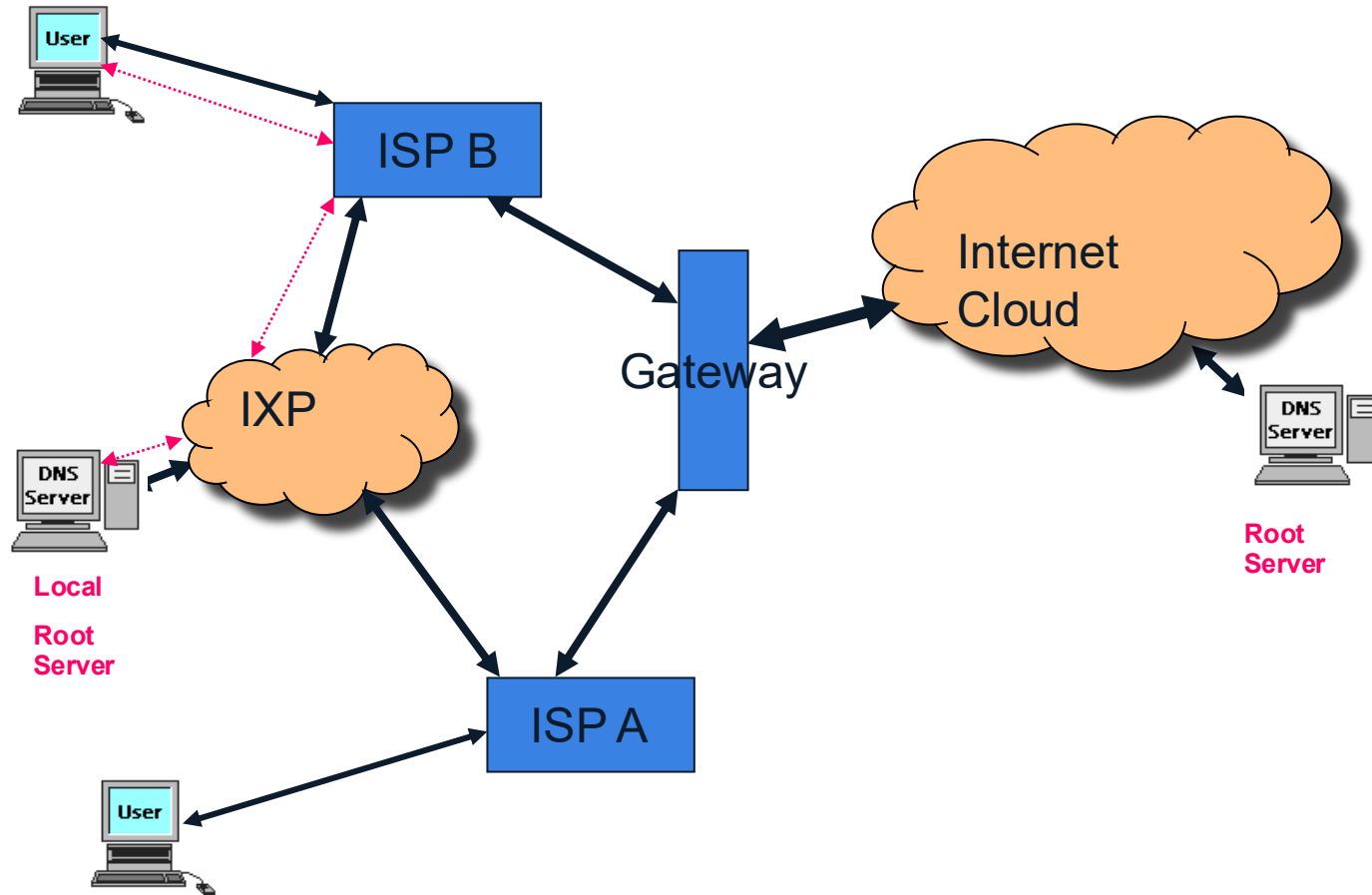


DNS Root-Server Instance: Before

- Most Internet services such as email and web depend on the DNS
- The Apex of the DNS is the Root
- Absence of a local Root Server means continued dependence on International connectivity for these Internet services



DNS Root-Server Instance: After



- **Autonomy** due to reduced dependence on International connectivity for local communications
- **Faster DNS** response times improve user experience
- **Improved resilience** encourages local hosting.



Polymakers enabling IXPs

By championing the development of IXPs and removing roadblocks to their growth and success

Foster relationships to learn more about local interconnection environments and sustainability models.

Learn from others through collaboration, training, and development opportunities.

Develop and enhance cross-border interconnection policies to build resilience between countries and regions.



We are here to help

ISOC Infrastructure, Technical communities and Community-led Networks supported (since 2020 – preliminary data)

- Development of new and existing IXPs
 - 64 supported
 - Over USD \$1,500,000 distributed through grants
- Technical Communities
 - 44 supported
 - USD \$200,640 distributed through grants



We are here to help

Internet Society
Foundation

[About](#) [Funding Areas](#) [Resources](#) [News & Stories](#) [Careers](#)

26 February 2026 | [Africa, Asia-Pacific, Europe, Foundation News, Latin America and Caribbean, Sustainable Peering Infrastructure](#)

Introducing the 2025 Sustainable Peering Infrastructure grantees

The 2025 [Sustainable Peering Infrastructure \(SPI\) Grant Program](#) has selected 24 grantees working to strengthen peering and interconnection in their local and regional contexts. To advance the [Internet Society's 2030 Strategy](#) to make access affordable and resilient for marginalized and underserved populations, the SPI program aims to improve local Internet interconnection infrastructure by establishing new, community-led [Internet Exchange Points \(IXPs\)](#) where they are needed, strengthening existing IXPs, and supporting the organizations and communities that enable peering and interconnection.

IXP Support

IXP Support grants fund the establishment of new IXPs where they are needed or enhance existing IXPs, helping them realize their full potential in the Internet ecosystem.

19 IXP Support grants were awarded in 2025:

Bangladesh Internet Exchange (BDIX) – Bangladesh

This project expands network capacity and extends connectivity support while aiming to bring in new members to increase the volume of local traffic exchange nationwide.

Bangladesh Internet Service Provider Internet Exchange Trust – Bangladesh

This project will establish a new IXP Point of Presence in the major urban center of Rajshahi. The project aims to reduce latency and costs, support universities and enterprises, and build a stronger regional peering community.



<https://www.isocfoundation.org/2026/02/introducing-the-2025-sustainable-peering-infrastructure-grantee/>

Training

We offer moderated online courses, face-to-face courses, and self-paced tutorials for the Internet Society community.

These are the most related with the topics above:

- Fundamentals of Designing and Deploying Computer Networks
- Introduction to Network Operations
- Advanced Network Operations
- Internet Exchange Points (IXP)
- Mutually Agreed Norms for Routing Security (MANRS)



But there's more!

Technical Skills

How to maintain Internet's technical infrastructure.

1. Designing and Deploying Computer Networks
2. Introduction Network Operations 1.0
3. Advanced Network Operations 2.0
4. MANRS
5. IXPs 2.0



Knowledge Exchange, Business, and Policy Skills

How to reduce the gap between policymakers and technical experts.

6. Community Networks
7. Internet Governance



Online Safety Skills

How to reduce the gap between policymakers and technical experts.

8. Digital Footprint
9. Privacy
10. Internet Security



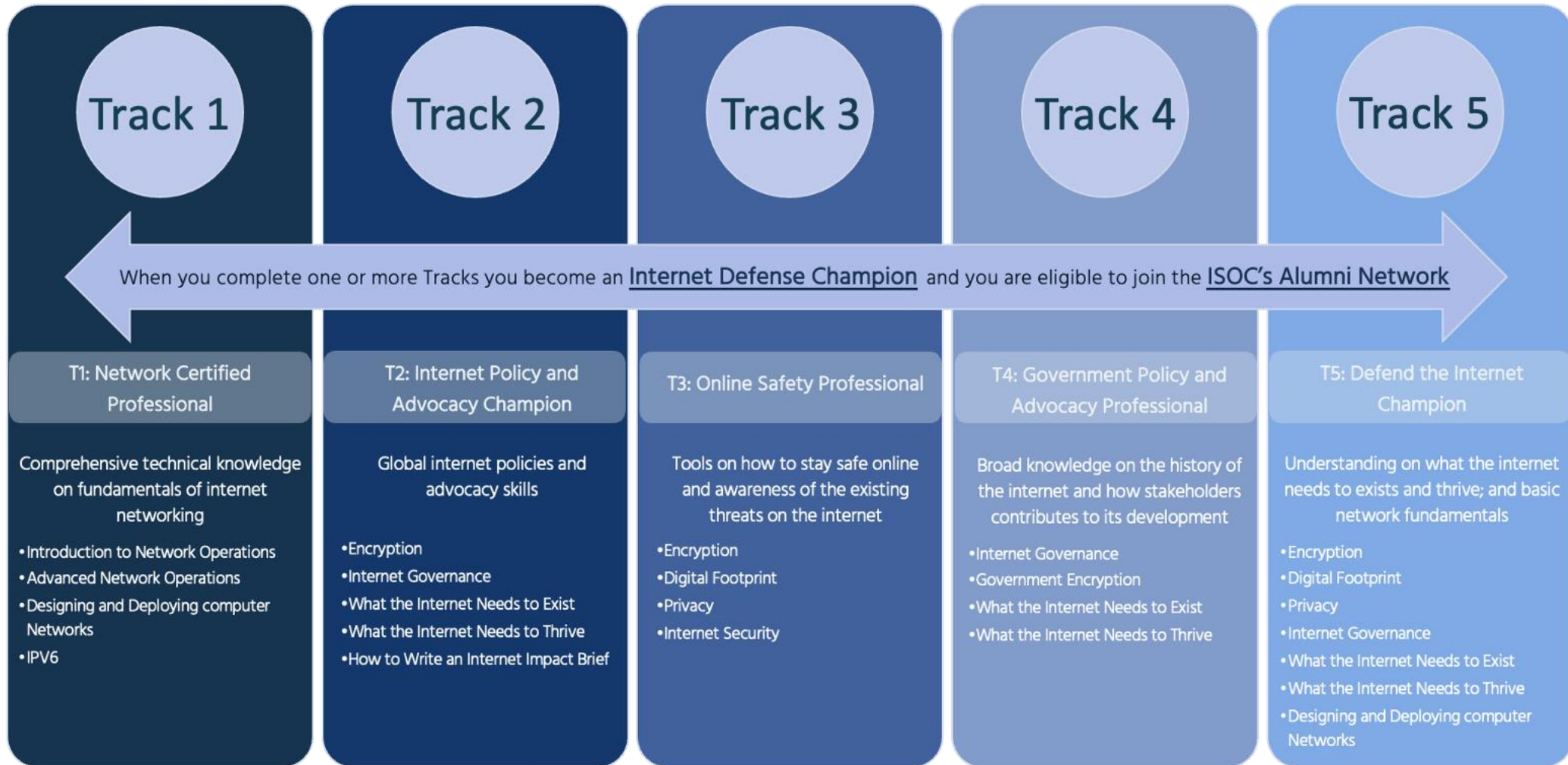
Advocacy Skills

How to champion protocols that keep the Internet secure.

11. Encryption
12. What the Internet Needs to Exist
13. What the Internet Needs to Thrive
14. How to write an Internet Impact Brief



Become an Internet Champion!



Peering Fellowships

Six-month program to equip 20 peering and interconnection professionals with skills, knowledge, and tools to address interconnection and peering challenges while maximizing opportunities that grow the Internet and develop the community at peering events..

What to expect:

- Online courses
- Attend an in-person peering event
- Advocacy training
- Engagement with industry professionals



<https://www.internetsociety.org/fellowships/peering>



The Internet is for everyone.

A small contribution can make a big difference.
100% of your donation helps to fund our work.



Thank you.



Acknowledgement and Attribution

This presentation contains content and information originally developed and maintained by the following organisation(s)



<http://www.internetsociety.org>



<http://www.euro-ix.net>